CLASS 7 — SCIENCE · FORMULA SHEET

Chapter 6: **Heat Transfer in Nature** | Quick revision, one page

Heat and temperature

- Heat is energy that flows from a hotter body to a colder body; it stops when temperatures become equal.
- Temperature tells how hot or cold a body is; the SI unit of temperature is the kelvin, and we commonly use degree Celsius (°C).

Modes of heat transfer

- Conduction: through solids from particle to particle; the particles do not move from their places.
- Convection: through liquids and gases by movement of the hot, lighter fluid upward and cold fluid downward.
- Radiation: through space without any medium (Sun to Earth); dark surfaces absorb more, light and shiny surfaces reflect more.

Conductors and insulators of heat

- Good conductors: metals such as copper, aluminium, iron (used for pans and radiators).
- Poor conductors (insulators): wood, plastic, air, wool, rubber (used for handles, clothes).
- Woollen clothes keep us warm as they trap air, a poor conductor.

Convection in nature

- Sea breeze: day time; land is hotter, warm air over land rises, cool air from the sea blows to land.
- Land breeze: night time; sea is warmer, so air blows from land to sea.
- Thermos flask: vacuum stops conduction and convection; silvered walls reflect radiation.

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